

Shirley Cai

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Research Interests

Health Economics, Labor Economics

Education

Emory University

Ph.D. in Economics

M.S. in Economics

Atlanta, GA

Spring 2027 (expected)

Summer 2026

Stanford University

M.S. in Statistics

B.A. in Economics

Stanford, CA

Spring 2022

Spring 2022

Job Market Paper

Medical School Environments and Physician Treatment Styles (with Ian McCarthy)

We examine the role of medical school environments in shaping physicians' later treatment decisions, focusing on a specific subset of cardiology in which physician discretion is particularly salient and separate from patient need. Using a year-matched measure of cath lab availability in the hospital referral region of each cardiologist's medical school, we estimate that a 10-percentage-point increase in training-period availability raises later catheterization rates by 0.58 percentage points, identified across graduation cohorts within medical school HRR. The imprint is concentrated among interventional cardiologists and operates through measurable features of the training environment rather than ranking or prestige. Combining this training imprint with effects from their peers or general practice environments, identified from mid-career movers, implies a long-run multiplier of roughly 1.5 on per-cardiologist cath rates. The findings suggest that training-side interventions, if pursued at scale, would propagate through cohort transition into a meaningful reduction in place-based variation in physician treatment decisions.

Working Papers

1. **Formation of Physician Referral Networks: Drivers and Potential Consequences** (with Ian McCarthy, Pablo Estrada, and Jillian Wilkins)

Works in Progress

1. **Nurse Scope of Practice Expansions and Physician Location**

Research Experience

Research Assistant to Matt Grennan, Emory University

Linked physician disciplinary action data to drug prescriptions and market exit.

Spring 2025 – Fall 2025

Research Assistant to Ian McCarthy, Emory University

Predicted physician race using images through the DeepFace library. Predicted physician race using name and location through the wru package. Evaluated name- and location-based predictions using external test sets.

Fall 2023 – Spring 2024

Research Assistant to Karen Eggleston, Stanford University

Explored relationships between quality of care and staffing outcomes in Japanese nursing homes. Headed analytical work adjusting for survey non-response bias.

Spring 2021 – Summer 2022

Research Assistant to Larry Goulder, Stanford University

Developed a model for pension fund dynamics fully coded and calibrated in MATLAB.

Summer 2020

Teaching Experience

Instructor , ECON 101: Principles of Microeconomics	<i>Spring 2026</i>
Teaching Assistant , ECON 372: Health Care Markets (Professor Ian McCarthy)	<i>Fall 2023 – 2024</i>
Grader , ECON 201: Intermediate Microeconomics (Professor Daniel Fershtman)	<i>Spring 2024</i>

Conference Presentations

Southern Economics Association 96th Annual Meeting , Houston, TX	<i>November 2026</i>
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Professional Service

Organizer , Applied Microeconomics Graduate Student Reading Group	<i>Fall 2024 – Summer 2025</i>
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Technical Skills

Software: R, Python, Stata, L^AT_EX